

EDUCATION

Doctor of Philosophy in Mechanical Engineering, GPA:3.88 Western Michigan University Research focus: Autonomous vehicles, Control and Optimization, Machine Learning and Deep Learning, Electrified Mobility	Kalamazoo, MI Aug. 2019 – June 2023
Master of Science in Mechanical Engineering, GPA: 4 Iran University of Science and Technology Thesis: Optimization of Electronic Throttle Modeling and Control for Adaptive Cruise Control System	Tehran, IR 2015
Bachelor of Science in Mechanical Engineering, GPA: 4 Semnan University	Semnan, IR 2012

RESEARCH EXPERIENCE

Postdoctoral Scholar <i>University of California, Davis</i>	Davis, CA Feb. 2024– Present
• Conducted research on Mobile DC Fast Chargers (MDCFCs) for Caltrans fleet electrification, including system design, cost modeling, and routing optimization using Python, MATLAB, and Google OR-Tools. •Supported hardware selection, procurement, and setup for mobile fast-charging systems for electric vehicles, including Keysight RP7900 series and FreeWire Boost Charger 200. •Assisted with Hardware-in-the-Loop (HIL) testing, power system integration, and safety validation for advanced energy and control systems at the UC Davis ATIRC Lab. •Led the instrumentation of electric vehicles (EVs) at the AHMCT Center with state-of-the-art sensors (GNSS, IMU, LiDAR) to enable self-driving control research, testing, and demonstration. •Coordinated field demonstrations and stakeholder meetings between UC Davis, Caltrans, and industry partners to evaluate EV charging and fleet electrification strategies. •Supported the development and mentoring of undergraduate and graduate researchers through technical guidance, skill-building workshops, and hands-on lab training. •Collaborated with Caltrans, NREL, and industry partners on technology evaluation, data analysis, and zero-emission fleet transition planning. •Authored and contributed to Caltrans technical reports, DOE VTO proposals, and peer-reviewed publications on co-design optimization of mobile DCFC systems for energy-efficient fleet operations.	
PhD Research Assistant <i>Western Michigan University</i>	Kalamazoo, MI Aug. 2019 – June 2023

Farhang Motallebiaraghi Curriculum Vitae

+1-269-861-0446

fmotallebi@ucdavis.edu



<https://www.linkedin.com/in/f-motallebi>

U.S. Department of Energy Project

Energy Efficient Autonomous Vehicles Enabled Through Infrastructure

- Measured, documented and compared lab's autonomous research vehicle sensors suite computation load and power consumption with basic version
- Developed ROS node for speed tracking on dynamometer in Python
- Calibrated chassis dyno using Argonne's National Lab database for UDDS and HWFET
- Planned coordination between collaborators for the dynamometer road load calibration
- Assisted with the management of the GitHub repositories for the project
- Developed a tutorial for data collection on the Lab's chassis dyno and initializing ROS nodes

Eco-driving Project in collaboration with CSU

Energy efficiency Improvements from Autonomous Eco-Driving with Traffic Light and Lead Vehicle Constraints

- Conducted literature review on eco-driving algorithms for automated electric vehicles • Assisted on the developing of eco-driving traces using Dynamic Programming, Genetic Algorithm and Particle Swarm Optimization in python
- Tested lab's research battery electric vehicle on chassis dyno with developed eco-driving traces and measured state of charge from CAN
- Compared CAN-derived state of charge from eco-driving traces with Simulation results from FastSim (NREL's vehicle powertrain simulator) for multiple drive cycles

TEACHING EXPERIENCE

Graduate Teaching Assistant *Western Michigan University*

Kalamazoo, MI
Aug. 2019 – June 2022

• ME 5950/6950 - Autonomous Vehicles

Spring 2020

This course takes a multidisciplinary approach by combining sensing, computer vision, artificial intelligence, controls, and systems engineering through the utilization of hands-on experiences with actual sensors and actual commercial vehicles. The software employed includes Python, ROS, and Carla.

• ME 4650 - Vehicle Dynamics

Spring 2022

Passive and active suspension and steering systems are designed for ride quality, handling, and safety. Python and MSC Adams are used to construct analysis techniques.

• ME 5950/6950 - Electric Vehicles

Spring 2022

In this project-based, cross-disciplinary course, students will learn to utilize Python to simulate and analyze electric vehicle performance.

• ME 3580 - Mechanism Analysis

Fall 2022

Analyses of position, velocity, acceleration, and force are carried out for a wide variety of mechanisms making use of graphical, analytical, and numerical techniques in addition to the applications of Solidworks, Python, and MSC Adams.

Farhang Motallebiaraghi Curriculum Vitae

+1-269-861-0446

fmotallebi@ucdavis.edu



<https://www.linkedin.com/in/f-motallebi>

• ME 2615 - Introduction to Mechanical Engineering

Fall 2019, Fall 2021

• ME 3350 - Instrumentation

Fall 2021

PEER REVIEWED JOURNALS

- 1. Deep Bilinear Koopman Model for Real-Time Vehicle Control in Frenet Frame**
Mohammad Abtahi, Farhang Motallebi Araghi, Navid Mojahed, Shima Nazari
IEEE Access, 2026. DOI: 10.1109/ACCESS.2026.3662607. [IEEE Xplore](#) [arXiv:2507.12578](#)
- 2. Real-Time Implementation Comparison of Urban Eco-Driving Controls**
Aaron Rabinowitz, Chon Chia Ang, Yara Hazem Mahmoud, Farhang Motallebi Araghi, Richard T. Meyer, Ilya V. Kolmanovsky, Zachary D. Asher, Thomas H. Bradley
IEEE Transactions on Control Systems Technology, vol. 32, no. 1, pp. 143–157, 2024. [IEEE Xplore](#)
- 3. Development and Evaluation of Velocity Predictive Optimal Energy Management Strategies in Intelligent and Connected Hybrid Electric Vehicles**
Aaron Rabinowitz, Farhang Motallebi Araghi, Tushar Gaikwad, Zachary D. Asher, Thomas H. Bradley
Energies, vol. 14, no. 18, art. 5713, 2021. DOI: 10.3390/en14185713. [MDPI](#)
- 4. Fleet Size and Mix Capacitated Vehicle Routing Problem with Time Windows for Mobile Fast Chargers**
Farhang Motallebi Araghi, Armin Abdolmohammadi, Navid Mojahed, Shima Nazari
Submitted to IEEE Transactions on Transportation Electrification, 2026 (under revision). [arXiv:2512.02381](#)

PEER REVIEWED CONFERENCES

- 5. Evaluation of Autonomous Vehicle Sensing and Compute Load on a Chassis Dynamometer**
Nicolas E. Brown, Farhang Motallebi Araghi, Johan Fanas Rojas, Shean Huff, Ben Feinberg, Zachary D. Asher
26th IEEE International Conference on Intelligent Transportation Systems (ITSC), 2023. [IEEE Xplore](#)
- 6. Autonomous Eco-Driving Evaluation of an Electric Vehicle on a Chassis Dynamometer**
Farhang Motallebi Araghi, Aaron Rabinowitz, Johan Fanas Rojas, Parth Kadav, Damon Miller, Thomas H. Bradley, Richard T. Meyer, Zachary D. Asher
SAE Technical Paper 2023-01-0712, 2023 SAE World Congress Experience. [SAE](#)
- 7. Road Snow Coverage Estimation Using Camera and Weather Infrastructure Sensor Inputs**
Parth Kadav, Nicholas A. Goberville, Kyle Prins, Amanda Siems-Anderson, Curtis Walker, Farhang Motallebi Araghi, Kyle Carow, Johan Fanas Rojas, Guan Yue Hong, Zachary D. Asher
SAE Technical Paper 2023-01-0057, 2023 SAE World Congress Experience. [SAE](#)
- 8. An Ultra-Light Heuristic Algorithm for Autonomous Optimal Eco-Driving**
Aaron Rabinowitz, Farhang Motallebi Araghi, Richard T. Meyer, Ilya V. Kolmanovsky, Thomas H. Bradley, Zachary D. Asher
SAE Technical Paper 2023-01-0679, 2023 SAE World Congress Experience. [SAE](#)
- 9. Mobility Energy Productivity Evaluation of Prediction-Based Vehicle Powertrain Control Combined with Optimal Traffic Management**
Farhang Motallebi Araghi, Kaisen Yao, Aaron Rabinowitz, Christopher Hoehne, Venu Garikapati, Jacob Holden, Eric Wood, Suren Chen, Zachary D. Asher, Thomas H. Bradley
SAE Technical Paper 2022-01-0141, 2022 SAE World Congress Experience. [SAE](#)